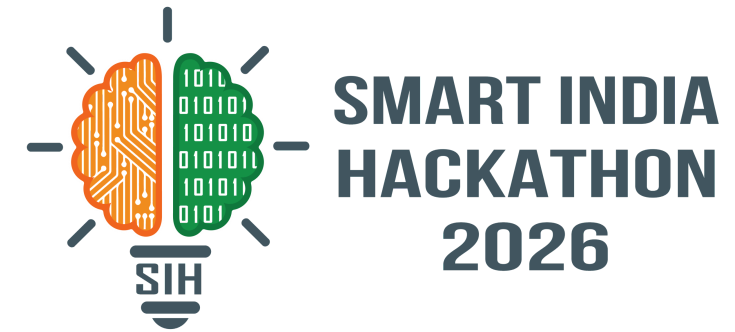




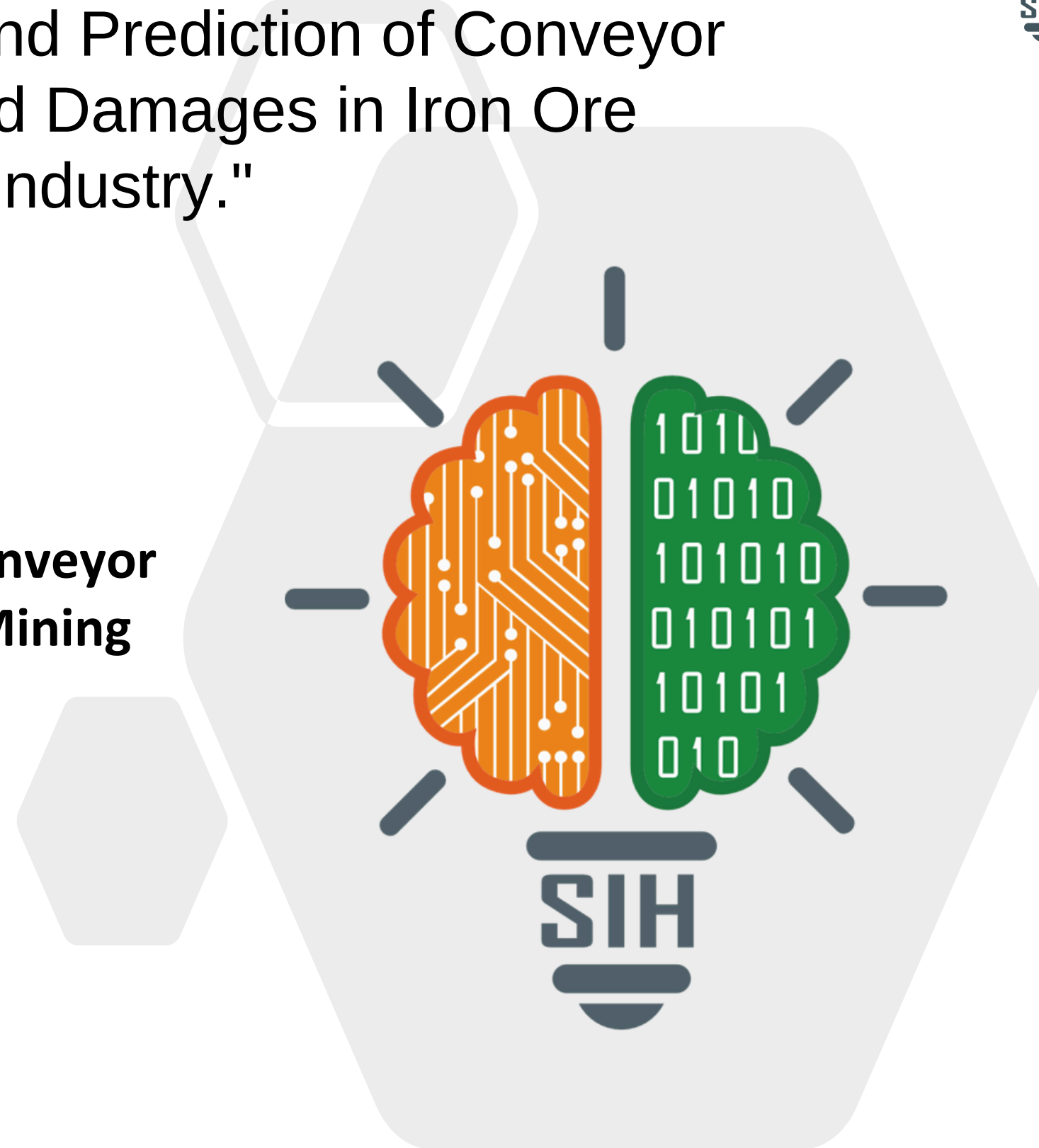
SMART INDIA HACKATHON 2026



" Intelligent Monitoring and Prediction of Conveyor
Belt Joint Rupture and Damages in Iron Ore
Mining Industry."

- **Problem Statement ID –SIH26008**
- **Problem Statement Title-**

"Intelligent Monitoring and Prediction of Conveyor
Belt Joint Rupture and Damages in Iron Ore Mining
Industry."
- **Theme-Smart Automation**
- **PS Category- Hardware**
- **Team ID-SIH26-A0H-T276**
- **Team Name (Registered on portal)-HexaCore**





CONVEYOR HEALTH MONITORING



SMART INDIA
HACKATHON
2026

Detailed Solution

To detect any abnormalities like more friction between conveyor belt and the rollers below, belt misalignment, and any jamming in the conveyor joint, power and torque delivered is measured.



To calculate motors power:

$$P = \sqrt{3}V_L I_L \cos \phi$$

The motors Torque is:

$$T = \frac{P}{\omega}$$



Since there will variations in torque and power according to the ore material on the belt, both parameters are measured in real time. So abnormalities will be identified instantaneously.



Torque allows to replace the mass of ore qualitatively in the equation so that the belt speed remains constant

Cameras are used to detect damage to belt. A camera with a top view from above and one from below.



The footage is then processed and any damage to conveyor belt is detected. If any, damage is evaluated and sent to dashboard.



Drones can also be used for detailed and remote inspection

Temperature

To ensure safety of the conveyor joint temperature is also measured besides power.

The sensor will measure temperature if power delivered goes above a threshold value. An alarm will be raised

What makes us different

Our system creates a normal motor-load signature for varying ore loads and detects unexpected increases in power/torque/current while maintaining constant belt speed.

So even if the ore load changes unpredictably, the system doesn't need to know the exact mass.

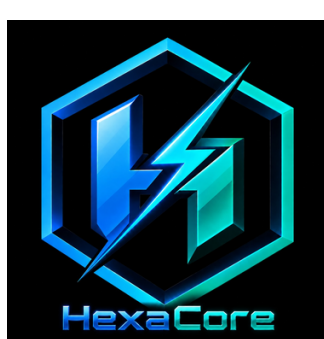
It asks:

“Given the current operating conditions, is the motor working harder than it normally should?”

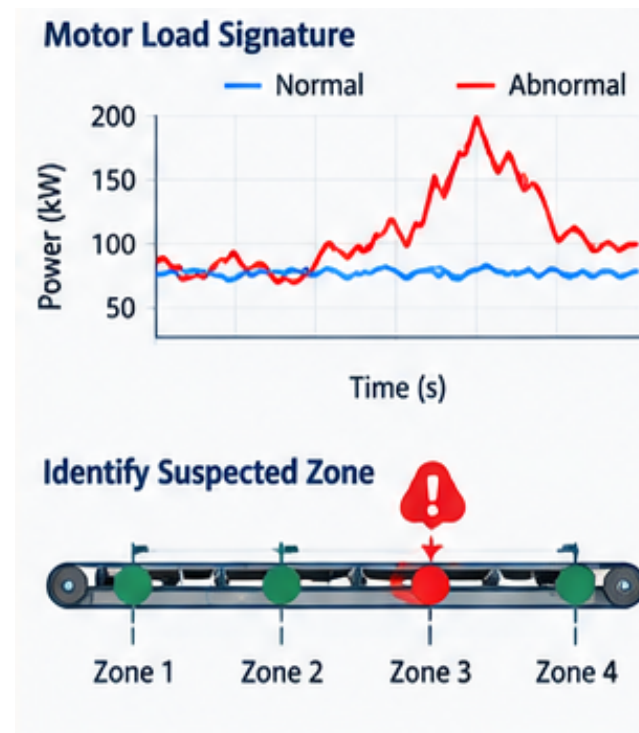
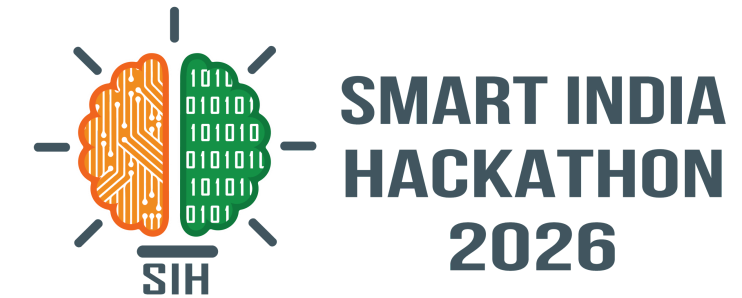
If yes → early anomaly alert → targeted inspection.

To find locations more precisely:

- We place vibration sensors at consecutive locations to detect the problematic location more precisely.



TECHNICAL APPROACH



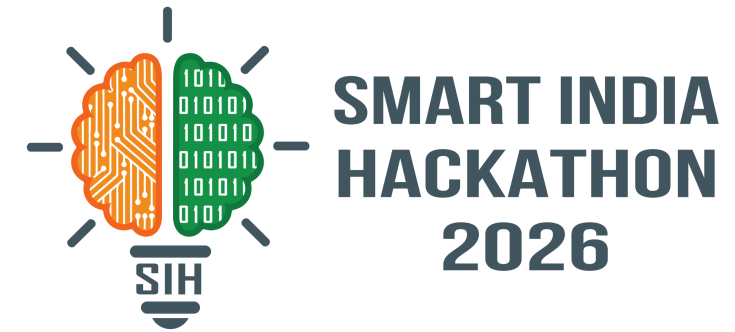
1. Sensors on the belt feel things — vibration, heat, misalignment.
2. IoT digitises the data, and transmits it
3. Those numbers go to a local mini-computer at the mine site.
4. That mini-computer runs the AI right there and figures out: "is this joint in danger?"
5. If yes, it sends that warning to the PLC (the machine that controls the belt) and the SCADA screen (the operator's usual screen) .
6. If it's really serious, a siren/light goes off on-site too, as backup.
7. Everything gets saved locally.



1. Industrial RGB cameras pixels (1920x1080 / 2 MP)
2. edge AI - YOLO + open CV
3. Constant speed control PLC+VFD
4. Real time monitoring
5. Torque Based Analysis
Expected vs Actual Torque
6. Mass independent AI
7. Fault Detection
8. Predictive Maintenance:
Generates early warning and trigger targeted RGB + thermal inspection for fault confirmation



FEASIBILITY AND VIABILITY



ECONOMIC

- Reduces downtime caused by unexpected belt and motor failures.
- Cuts maintenance costs through early fault detection.
- Prevents production losses by enabling timely maintenance
- Uses existing PLC/VFD systems, reducing installation costs.
- Improves return on investment (ROI) through continuous monitoring.
- Extends component life, reducing frequent replacement costs.

The system can be integrated with existing PLC/VFD infrastructure, making it relatively low-cost and scalable across multiple conveyor sections. The initial investment can be recovered through reduced downtime and preventive maintenance costs.

TECHNICAL

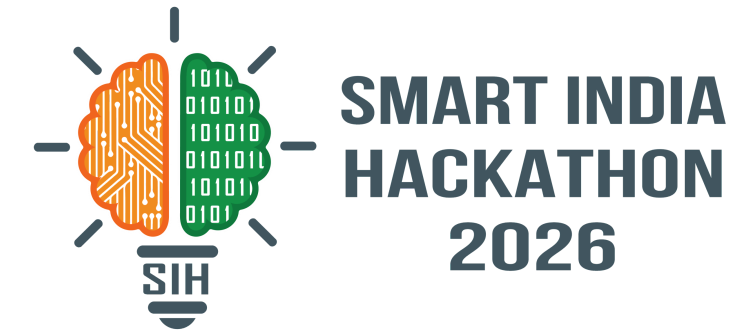
- Constant-speed operation — VFD + speed feedback maintains the conveyor at a fixed belt speed despite continuously varying ore load.
- Real-time load indication — Motor current rises with increasing torque demand; no direct mass measurement is required.
- Fault detection — Persistent deviation between predicted and measured power/torque indicates abnormal friction, roller/bearing failure, belt misalignment or possible splice damage.

SOCIAL

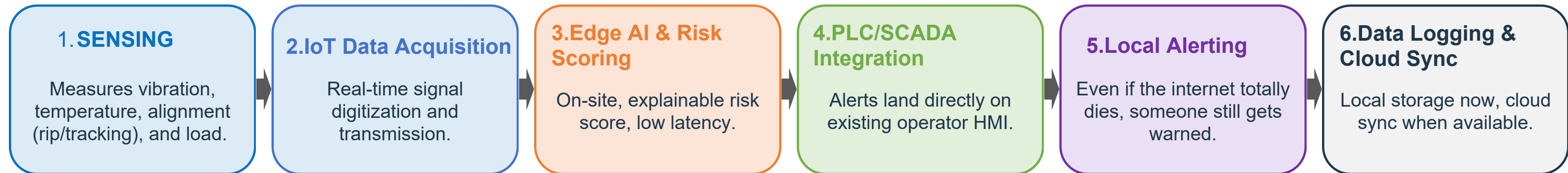
- Supports faster emergency response through real-time alerts.
- Creates opportunities for skilled technical jobs in AI and automation.
- Reduces worker exposure to dangerous conveyor-belt inspection and maintenance.
- Early-warning alerts let operators act before a rupture becomes serious.
- Helps prevent production interruptions.



IMPACT AND BENEFITS



IMPACT FLOW



KEY BENEFITS

- Fewer unplanned breakdowns**

Catches a weakening joint early, before it actually fails.

- Works without reliable internet**

Runs fully on-site, so it stays effective even at remote, low-connectivity mine locations.

- No need to replace existing equipment**

Integrates directly with whatever PLC/SCADA system the mine already uses.

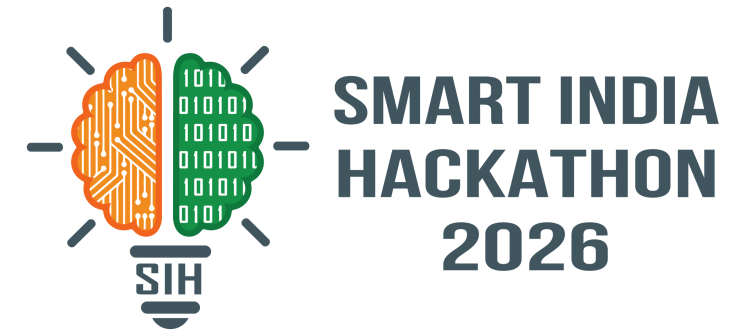
- Builds operator trust**

Every alert comes with a clear reason behind it, not just a raw number.

Prototype success can be demonstrated through: continuous sensing • Edge AI Digitalisation • Control Room Integration • Local & Cloud Sync



RESEARCH AND REFERENCES



- Author(s), Title of paper, Journal/Conference, Year.
- Author(s), Title of paper, IEEE/Elsevier, Year.
- CEMA, Belt Conveyors for Bulk Materials.
- Directorate General of Mines Safety (DGMS), Government of India, relevant mining safety regulations.
- Indian Bureau of Mines, Government of India, mining statistics/reports.
- AI Models-Claude and Gemini